Weigh-at-Source Kit — Weighing Concept (Part 1)

Nov 6, 2026 · @Minh Le

Summary

We propose weighing the container directly with four stainless shear-beam load cells under its support points, read by a 24-bit weighing module that plugs into the existing SMART controller. Target: σ ⩽ 1.5 % of load, which keeps 97 % of 2.5–3.5 t loads within ±100 kg in our sensitivity simulation (challenge target ≈ 95 %).

Two design choices make it robust and cheap to run:

- **Read only when the press is idle.** The controller already knows when the ram moves, so dynamic forces never reach the reading.
- **Two separate zeros, never a blind re-tare.** The load cells are zeroed while the frame is empty (between pick-up and set-down), which removes sensor drift. The container's own weight comes from a register keyed by an RFID tag on each container, and a set-down reading updates it only when the container is identified, reported empty and within tolerance; residue, a swapped or unknown container is flagged instead of becoming the new zero.

Hydraulic pressure and motor current stay as secondary signals (state detection, anomaly checks), not as the mass source: compaction pressure reflects how hard the material resists, not how much it weighs.

The kit fits DUO units as a bolt-on base frame under the detachable container; COMBI units need weigh pads under the unit's feet. Estimated cost: about €2.0–4.6k hardware plus half a day of installation at one-off list prices, lower in series and when factory-fitted. Not yet tested on a real Europress machine; the validation plan below is how we would prove it.

Requirements

The accuracy a sensor needs depends on the use case: €/t reporting tolerates a few percent, but checking an operator's invoiced tonnage needs σ of about 2 % or better. This is a **sensitivity simulation, not a hardware test**: we re-measured 5,787 clean weighbridge tickets from the synthetic dataset (single load, vehicle tare actually weighed, so no stored-tare bias) with each candidate error model, using the method of `examples/weighing_accuracy_value.py`. Whether real hardware reaches the needed σ is what the validation plan below must show.

Sensor σ (% of load)	2.5–3.5 t loads within ±100 kg	Detects +10 % invoiced tonnage	Median €/t error
1.5 %	97 %	95 %	1.1 %
2.5 %	84 %	92 %	1.7 %
5 %	53 %	47 %	3.3 %
10 %	27 %	16 %	6.8 %

From the brief and the data we derive five requirements:

1. **Accuracy:** σ ⩽ 1.5–2 % in normal operation (challenge target: ±100 kg on a 3 t load for about 95 % of readings).
2. **Affordable at fleet scale:** standard parts, no per-unit engineering, cost low enough to pay back from the services it enables (see Cost below).
3. **Retrofit-friendly:** installable on site in about half a day, without modifying the hydraulics.
4. **Robust:** IP68, −30 °C winters, impacts when a container is set down, no drift between lifts.
5. **Trustworthy:** every reading carries a quality status, and drift is detected automatically.

Sensing options

Load cells are the only option that meets the accuracy requirement without per-site learning; the others are kept as secondary signals.

Option	Expected accuracy	Hardware cost	Retrofit	Main weakness	Role
4 shear-beam load cells under the container supports	≈ 0.5–1.5 % (mechanics-limited)	≈ €2.0–4.6k hardware (list)	bolt-on base frame (DUO), weigh pads (COMBI)	debris/ice under the frame, side loads	**Primary mass source**
Weld-on strain sensors on the frame or rails	≈ 2–5 % after per-unit calibration	lowest	welding on site	zero drift, temperature, per-unit calibration	Low-cost variant to test
Hydraulic pressure + ram stroke	> 10 %, material-dependent	none (already sensed)	software only	pressure = resistance to compression, not mass	State detection, anomalies

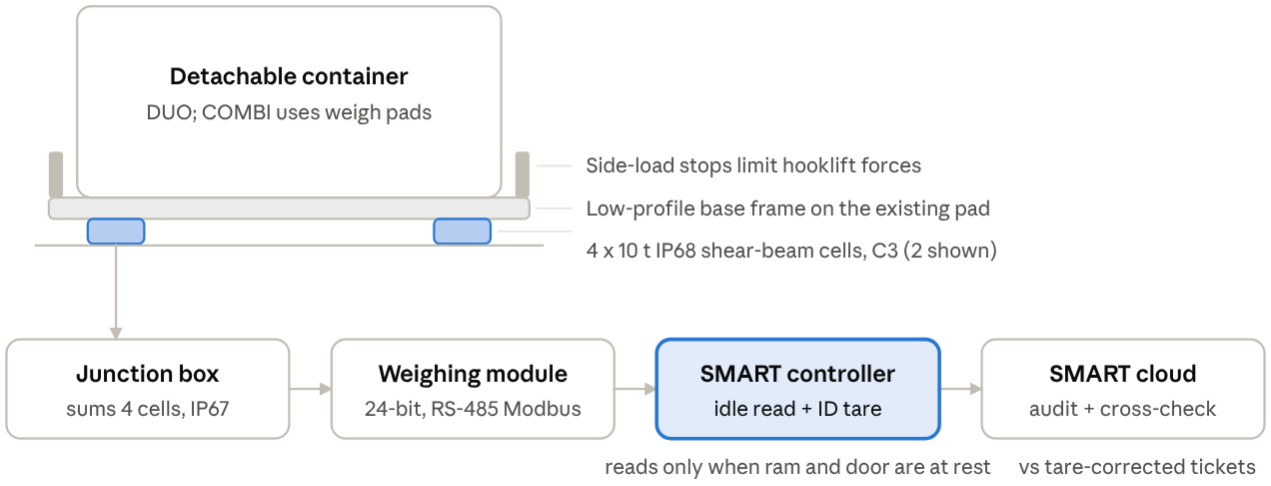
Option	Expected accuracy	Hardware cost	Retrofit	Main weakness	Role
Motor current	> 10 %	none	software only	same as hydraulics	State detection
Camera at the hopper	not a weight sensor	moderate	needs light, cleaning	dirt, darkness, privacy	Contamination evidence (future)

The accuracy figures for strain sensors, hydraulics and current are our engineering estimates, not measurements; the validation plan tests the strain-sensor variant against load cells on the same rig.

Recommended design

The container's weight is carried by four load cells, and the existing SMART controller decides when a reading is valid. We use one part for all DUO sizes: four 10 t C3 cells, sized against the largest container in the dataset (30 m³, up to 16.7 t of material) in the load check below.

Four load cells under the container feed the existing SMART controller



Weigh-at-Source Kit, DUO retrofit · placement and signal chain (schematic, not to scale)

The container rests on a base frame carried by four load cells; the signal is summed, digitised and read by the SMART controller only when the press is idle, then cross-checked in the cloud.

DUO (stationary compactor, detachable container). A low-profile steel base frame sits on the existing pad where the container is parked. Four shear-beam cells carry the frame's rails at the container's support points. Mechanical stops limit side loads when the hooklift drops or pulls the container, and the cells sit in overload-protected mounts.

COMBI (compactor and container joined). The whole unit is lifted away, so cells cannot travel with it cheaply. We use four weigh pads under the unit's feet at its parking position; the controller subtracts the unit's own tare, which is learned automatically (see Calibration).

Signal chain.

1. Four 10 t IP68 shear-beam cells (C3 class) with 6-wire cables, plus a UHF RFID antenna that reads the container's tag.
2. Stainless junction box summing the cells; a 4-channel module keeps each cell visible for diagnostics.
3. 24-bit weighing module (RS-485 Modbus RTU) mounted in the controller cabinet, powered from the existing 24 V supply.
4. SMART controller polls the module and sends mass with the existing telemetry.

Measurement logic in the controller.

- **Idle reading:** take the median of 20 samples only when the ram is at rest and the door is closed; never during compaction.
- **Sensor zero:** when the frame is empty (container lifted away, readings stable for 60 s), record the frame-only reading; accept it if within ±50 kg of the frame's commissioning value, otherwise flag a cell or debris problem.
- **Container tare:** from the container register (next section), never from an unchecked set-down reading.
- **Collected mass per lift** = last idle gross reading before pick-up − sensor zero − registered tare of the departing container. Material generated in the cycle = collected mass − any residue the container brought in.
- **Quality flags:** cell imbalance, sudden zero jumps, unknown container and temperature outside the calibrated range mark a reading as degraded.

Container identity and tare

On a DUO site the container that comes back is often not the one that left, and it may still hold material, so the tare must belong to the container, not to the site.

- **Identity:** a UHF RFID tag (IP69K) on each container, read by an antenna on the frame at set-down. Fallback: the driver or site scans the container's QR code in the SMART app.
- **Register:** SMART keeps one tare per container ID, first set when the container is weighed empty (at installation, at the depot, or from an operator ticket with a weighed tare).

Set-down situation	What the controller does
ID known, fill sensor < 5 %, stable, within ±max(30 kg, 1 %) of the registered tare	accepts; the register follows slowly (moving average), so one odd reading cannot move it
ID known, reading above the band	residue: tare unchanged, the excess is recorded as carried-in mass of the new cycle and flagged
ID known, reading below the band	damaged or modified container, or a sensor fault: reading degraded, service check
No tag read or new container	no tare update; volume-class tare used and the mass marked low-confidence until the container is weighed empty

COMBI units are one machine with a fixed ID, so only the sensor-zero and plausibility rules apply.

Load and impact check

Four 10 t cells carry the worst static case at 89 % of their rating; the critical case is a hooklift picking up a full container, where the two rear supports briefly carry almost everything, so mechanical overload stops take any load above about 110 % of rating into the frame. Container tares are taken from public listings: 2.4–3.0 t for open 20 m³ hooklift containers (German Distribution) and 4.4–5.6 t for compactor containers (example listing); we assume up to 5 t for the largest DUO container, to be replaced by Europress drawings.

Load case (30 m³ DUO)	Total load	Worst support	Per 10 t cell	Protection
Full, static, material pushed to the rear (40 % on one support)	5 + 16.7 + 0.5 frame = 22.2 t	40 %	8.9 t (89 %)	within rating
Hooklift pick-up of a full container (front lifted)	22.2 t	2 rear supports	11.1 t (111 %)	overload stops engage at about 110 %; cell safe overload 150 %
Empty container set down hard (impact factor 3, assumption)	5 t × 3 = 15 t	2 supports	7.5 t (75 %)	within rating

Load case (30 m³ DUO)	Total load	Worst support	Per 10 t cell	Protection
Hook pulling or pushing sideways	horizontal	—	kept below 10 % of rating	side stops, rocker mounts

The impact factor of 3 and the 40 % share are assumptions; the rig test measures both with high-rate logging. With a typical ultimate overload of 300 % for shear-beam cells (to be confirmed on the chosen datasheet), even the pick-up case without stops keeps a safety factor above 2.5.

Indicative bill of materials and cost

A DUO retrofit costs roughly €2,450–5,200 installed at one-off list prices; series purchasing and a frame designed for production should lower it substantially, which a supplier quote must confirm. Component prices are single-unit list prices we found online (converted at about £1 = €1.15 and $1 = €0.90); everything marked *estimate* is our assumption.

Item	Qty	Unit cost (€)	Total (€)	Basis
Stainless IP68 shear-beam load cell, C3, 10 t	4	200–450	800–1,800	0.5–2.5 t class lists from about £135 ex VAT (Dini Argeo SBX); 10 t estimated higher, to be quoted
Mount with overload and side-load stop	4	50–150	200–600	estimate
Summing / junction box, IP67	1	80–150	80–150	lists at $89 (Laumas CE41N)
4-channel 24-bit weighing module, RS-485 Modbus	1	260–550	260–550	$289–350 (Laumas TLB4); $590 (Transcell LCT-2)
UHF RFID reader + antenna on the frame	1	150–400	150–400	estimate
RFID tag per container (IP69K)	1–3	5–20	5–60	estimate; one per container in rotation
Low-profile steel base frame (DUO)	1	400–900	400–900	estimate

Item	Qty	Unit cost (€)	Total (€)	Basis
Cables, glands, cabinet parts	1	50–100	50–100	estimate
Hardware			**≈ 1,950–4,550**	
Installation: 2 technicians, about half a day	1	500–650	500–650	estimate

COMBI weigh pads replace the frame and mounts; their cost is similar and also needs a quote. Integrated into new machines at the factory, the frame and installation largely disappear, so the marginal cost is mainly the cells and the module.

Installation and retrofit

A DUO retrofit is planned to take two technicians about half a day, timed with a scheduled lift so the container is away. These times are estimates to verify in the pilot.

1. **Survey (remote, from SMART data):** container size, pad condition, controller model, free RS-485 port and 24 V capacity.
2. **Prepare the pad:** clean and level the parking position; check flatness (shims under the frame feet if needed).
3. **Set the frame:** place the pre-assembled base frame with cells and stops; align it with the hooklift guide rails; anchor it.
4. **Wire:** cells to the junction box, junction box to the weighing module in the controller cabinet; module on the 24 V supply and RS-485 bus.
5. **Update the controller:** load the idle-reading and auto re-tare logic over the air; enable mass in the telemetry message.
6. **Commission:** record the empty-frame zero, tag every container in rotation and register its empty weight, then run the calibration below before the first loaded reading counts.

COMBI: the same steps with weigh pads at the unit's feet instead of a frame. New machines: cells and module mounted at the factory; only steps 5–6 on site.

Calibration

Calibration happens once at installation with a known load and then continuously against operator weighbridge tickets, so a drifting unit is found within a few lifts.

At installation.

- **Span:** place a known load in the container, for example a 1,000 L IBC tank filled with water metered by a flow meter (≈ 1,000 kg), or certified test weights. Record the reading, then repeat with the load in each corner (corner test) to check cell balance. Because 1 t is far below the design range, add a high-load span check: a loaded container weighed on a certified weighbridge just before and after it is set on the frame.
- **Tare:** record the empty-frame zero, then weigh each tagged container empty and enter it in the container register; COMBI units store the unit's own tare.
- **Record:** calibration factor, corner errors, temperature and installer in the SMART device record.

In operation.

- **Sensor zero whenever the frame is empty** removes load-cell drift without touching any container's tare.
- **Container tare register** with the acceptance rules above; a tare is only learned from an identified, empty container.
- **Weighbridge cross-check:** when the operator shares a ticket, compare the device's collected mass with the ticket's net. Before comparing, correct the ticket for stored vehicle tares and split combined loads: in the dataset, a truck's stored tare differed from its weighed tares by up to about 230 kg, which would otherwise be blamed on the sensor (our solution's `weight_bias.py`).
- **Drift alarm:** a rolling median of device ÷ corrected ticket outside 1 ± 3 % over the last 10 shared lifts flags the unit for recalibration.
- **Cell health:** a growing imbalance between the four cells signals a damaged cell, debris or ice under the frame.

Error budget

The estimated combined error on the net material mass is σ ≈ 1.2 % of a 3 t load (≈ 35 kg) in a typical case and ≈ 1.9 % (≈ 57 kg) in a pessimistic one. In the sensitivity simulation, σ = 1.2 % keeps about 99 % of 2.5–3.5 t loads within ±100 kg, but σ = 1.9 % only about 92 %, below the 95 % target. The two levers that decide it are the mounts (side forces) and the quality of the container tare register; the rig test measures both. All values are engineering estimates, combined as the root of the sum of squares, to be replaced by rig measurements.

Error source	Typical σ (% of 3 t load)	Pessimistic	Mitigation
Side forces and friction in mounts	0.7	1.0	side-load stops, rocker mounts
Off-centre load, mounting non-linearity	0.5	0.8	four-cell summing, corner test, full-range rig test
Debris or ice bridging the frame	0.5	0.8	cell-imbalance flag, open frame design, empty-frame zero check
Container tare from the register (identified, verified empty)	0.5	1.0	RFID identity, acceptance band ±max(30 kg, 1 %), slow moving average
Installation calibration	0.3	0.5	metered water load, corner test, high-load span check
Load cell combined error (C3, 4 × 10 t)	0.13	0.17	—
Temperature effect on span (−30 to +30 °C)	0.07	0.1	temperature-compensated C3 cells
ADC noise, 24-bit	0.01	0.01	median of 20 idle samples
Combined (root sum of squares)	**≈ 1.2**	**≈ 1.9**	
Not in the combined figure: unknown container, volume-class tare	≈ 5 or more	—	marked low-confidence; never used to verify an operator's tonnage

Zero drift is not in the table because the sensor is re-zeroed every time the frame is empty. A wrong container tare would be a systematic error, which is why tares come only from identified, empty containers and unknown containers are marked low-confidence. Not sensor errors, but visible as differences against the weighbridge: material added between the last reading and the lift, residue left in the container, and the weighbridge's own error (about 12 kg in the synthetic world).

Validation plan

The concept is proven in two steps: a rig test that measures the error budget, then a field pilot that compares the device with operator weighbridges. It passes if ≥ 95 % of normal-state readings above 1 t fall within ±3.3 % of the corrected weighbridge net.

Step	Set-up	Measures	Pass criterion	Duration
1. Rig, static (low range)	one frame + container at Europress; test loads of 250–4,000 kg in 5 positions	linearity, corner error, repeatability	σ ≤ 0.5 %	1 week
2. Rig, full-range linearity	hydraulic load frame or certified weights at 1, 5, 10, 15 and 20 t total; off-centre loads up to 9 t on one support	linearity, span and corner error across the design range	error ≤ 0.5 % of reading at every step; corner error ≤ 0.5 %	1 week
3. Rig, load and impact	proof load of 125 % of rating on each support; 1,000 hooklift pick-ups and set-downs with high-rate logging; side pulls	peak loads, real impact factor, zero after each cycle	no zero shift > 0.1 % after proof load; stops engage before 110 %; impact factor confirmed or cells resized	2 weeks
4. Rig, environment	outdoors or cold room, −25 to +25 °C	temperature effect on span and zero	span change ≤ 0.2 %	2–3 weeks
5. Rig, tare logic	swap between 3 tagged containers, one with 200 kg residue, one untagged	correct tare chosen, residue and unknown container flagged; spread of repeated empty-tare readings	100 % of swaps handled as the table prescribes; tare spread σ ≤ 15 kg	1 week

Step	Set-up	Measures	Pass criterion	Duration
6. Rig, variant	weld-on strain sensors on the same frame, read in parallel	low-cost variant vs load cells	decides whether the cheap variant meets σ ⩽ 2 %	in parallel with 1–4
7. Field pilot	20–30 units (DUO and COMBI, cardboard, mixed waste, biowaste), customers whose operator shares tickets	device vs tare-corrected ticket, per lift	⩾ 95 % within ±3.3 %; drift alarms ⩽ 1 per unit per quarter	2–3 months

The dataset's simulation assumes this accuracy; the pilot is what would show it is achievable.

Limitations and untested assumptions

The device weight would be a verification basis, not a billing basis: legal-for-trade use would need type approval as a non-automatic weighing instrument and periodic verification, which this concept does not attempt.

- **Untested on real machines.** No Europress drawings, controllers or test rigs were available; mounting points, controller ports and power budget are assumptions.
- **Costs are list-price estimates.** Supplier quotes for 5–10 t cells, mounts and a production frame are needed.
- **COMBI is harder.** Weigh pads at a fixed parking position assume the unit always returns to the same spot.
- **Accuracy figures are engineering estimates** until the rig test replaces them.
- **Synthetic evidence.** Requirement figures come from the Since AI synthetic dataset; they show which accuracy matters, not that it is achieved.
- **Data rights.** Weighbridge cross-checks depend on operators sharing tickets; without them, calibration relies on the installation check and cell-health monitoring.
- **Container identity is a precondition.** Without a readable tag (or a QR scan), a swapped DUO container gets only a low-confidence class tare; every container in rotation needs a tag and one empty weighing.
- **Loads and impacts are calculated, not measured.** Container tares come from public listings and the impact factor is assumed; the rig's load test decides the final cell size.